Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name	**Light Gravity Bot**
Team leader	**Yawei Chen**
Team member	**Pratyush Gupta**
Problem statement	**AI output needs a work record, evidence and a human acceptance decision.**

Brief about the idea

AI work that a team can inspect and accept

For individuals and small teams turning source material into software, documents and everyday deliverables.

THE TEAM'S PROBLEM

Context scatters across chat.
Completion still needs evidence.

LIGHTGRAVITY'S ANSWER

Each task binds the goal and file, checks, review and human decision.

Future of Work & Enterprise Productivity · Shared-workspace MVP · File records + Task Ontology

Opportunities

The handoff is the product

Work pattern	What the next teammate needs
Chat alone	Reconstruct the goal, context and latest answer
A generated file	Find its inputs, checks and approval status
LightGravity task record	Open the candidate with its evidence and decision

INITIAL USE CASE

Small development and source-based delivery tasks

PILOT SUCCESS MEASURES

Time to acceptance · rework rounds · evidence completeness

List of features offered by the solution

PEOPLE SHARE THE WORKSPACE

Individual identity

Admin/member accounts. Personal chat is owner-only; Team work stays shared.

Intentional invocation

Ordinary Team messages stay quiet. An explicit @bot invokes the assistant.

Human control

Inspect the file and review record, then accept or request a change.

AI TEAMS FORM AROUND A TASK

275 recruitable role templates

Goal-fit metadata selects specialists with supported tools and configuration.

Application-managed execution

Plan validation and dependency scheduling govern task-scoped workers.

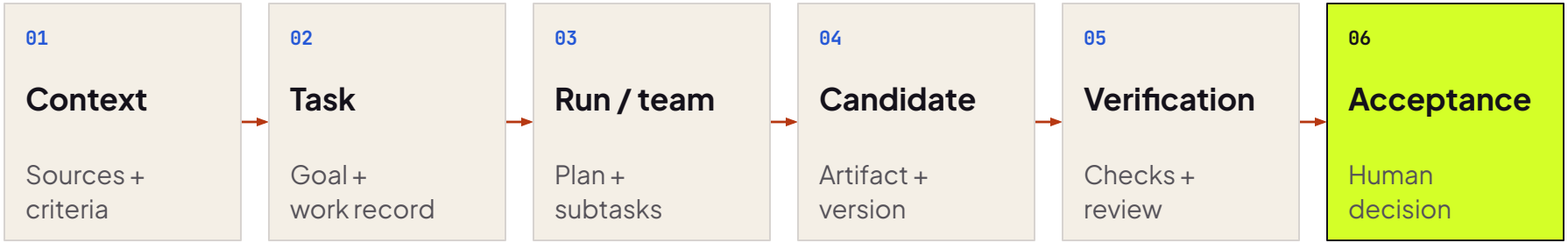
Independent review

A separate context inspects the candidate. Unavailable review blocks acceptance.

Role templates are capabilities, not simultaneous workers. Recruitment is deterministic; autonomous swarm remains a research direction.

Process flow diagram or Use-case diagram

One task. A traceable delivery lifecycle.



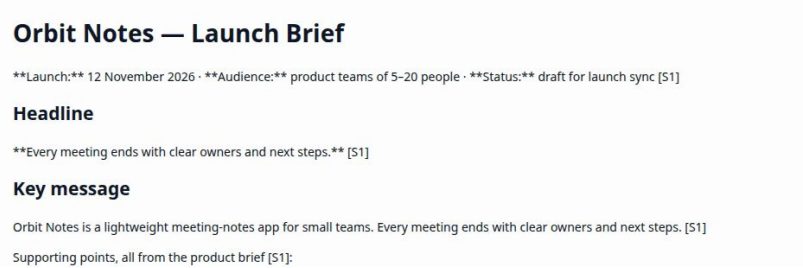
Rework changes the candidate. Acceptance stays tied to the reviewed snapshot.

Task Ontology: explicit objects, stable references and program-validated transitions. General knowledge-graph reasoning is outside scope.

Wireframes/Mock diagrams of the proposed solution (optional)

SUPPLIED LAUNCH WALKTHROUGH / TAKE 5

A launch brief a teammate can inspect



TEAM HANDOFF

Ben opens the brief

Versioned source context
Source-referenced output
Review before acceptance

Cropped from supplied Ben recording. Launch brief passed review; human acceptance remains pending.

Real delivery example

A cheaper proposal can still fail the brief

12 seats. Mandatory SSO, EU data region and CSV export.

Vendor	First-year cost	Requirement coverage
Cedar Support	$2,892	SSO, EU region and CSV included
Maple Desk	$2,016	SSO missing; EU region unspecified

Recommendation: Cedar, with unresolved contract questions

Synthetic supplier task. Source-derived arithmetic and recommendation manually checked. Human accepted after separate model review.

Architecture diagram of the proposed solution

EXECUTION PROPOSES. APPLICATION CODE GOVERNS.

Workspace + API	React / TypeScript → FastAPI / identity
Plan + scheduler	Validate tools, connections and dependencies
Agent execution	Antigravity SDK → configured generation provider
Verification	Authorized checks + independent reviewer context
Work record	Markdown / YAML / JSONL + run-scoped artifacts

THREE ENFORCEMENT BOUNDARIES

01 **Permission**

A plan cannot add authority.

02 **Evidence**

Owning Run + matching digest.

03 **Acceptance**

Missing review is not a pass.

Deploy path: GitHub OIDC → Artifact Registry → Cloud Run. Current container-local state requires durable storage before scale-out.

Technologies to be used in the solution

Google AI in the execution path

Technology	Role in LightGravity	Evidence in this demo
Antigravity SDK 0.1.22	Agent sessions and scoped tools	Real supplier file produced through SDK
Generation providers	Coordinator, worker and reviewer	Gemini supplier run; DeepSeek launch run
Gemini native audio	English machine transcription	Synthetic customer call transcribed
EmbeddingGemma 2	Semantic retrieval over sources	Indexed in supplied launch recording
Cloud Run + FastAPI / React	Serve the shared team workspace	Hosted contract 1.26.0 observed

Supplier: Gemini. Supplied launch: DeepSeek worker/reviewer, Gemini audio and EmbeddingGemma indexing.

Estimated implementation cost (optional)

PROTOTYPE ECONOMICS / MEASURED NEXT, NOT ASSUMED

Cost per accepted task is the useful metric

Cost component	What determines it	Current evidence
Model inference	Tokens, calls, retries and review	Live Board reports usage unknown
Cloud Run hosting	CPU / memory allocation and active time	Hosted service; bill not audited
Storage and operations	Sources, builds, registry and retention	Needs a durable-storage pilot

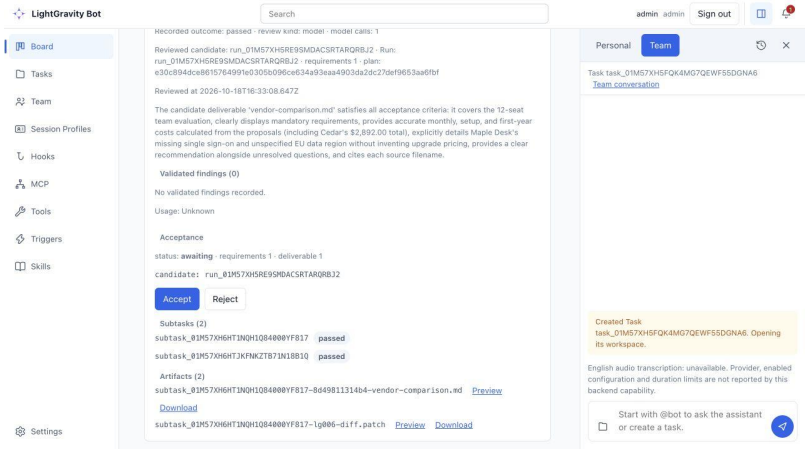
Pilot calculation: total model + hosting cost ÷ accepted tasks

Usage accounting is not a hard dollar spending cap. No observed ROI or production cost claim.

Snapshots of the prototype

HOSTED PROOF / INDEPENDENT MODEL REVIEW

The supplier task reached human acceptance



2 file checks passed

Separate reviewer passed

Human accepted the verified comparison

Checks validate evidence presence and artifact location. They do not prove every sentence or citation is correct.

Prototype Performance report/Benchmarking

RECORDED WORKFLOWS / 18 OCTOBER 2026

Two walkthroughs and their recorded outcomes

Evidence shown	Verified outcome	Acceptance
Supplier comparison	Source-bound file, checks and model review passed	Accepted
Launch brief + sources	PDF, OCR, audio and draft; review passed	Awaiting acceptance
Teammate inspection	Ben opens the source-referenced launch file	Approval pending

Human control: acceptance stays a separate decision

Supplier accepted. Launch review passed. Human approval pending.

Additional Details/Future Development (if any)

TOWARD OLAV-OS / STAGED EXPANSION

Scale the work model, then the infrastructure

NOW

Bounded shared workspace

Task teams, file records, review and human acceptance

NEXT

Reliable pilot delivery

Explain failures, expose usage, validate persistence and recovery

LATER

Broader team workflows

Finer permissions, Office artifacts and governed human / agent work

Evaluate GCS + lakeFS for versioned artifacts. Multi-process admission and conflict policies are prerequisites for broader concurrency.

Provide links to your:

SOURCE REPOSITORY

github.com/yhvh-chen/lightgravity-bot

Frozen source inspected at 724eb0e. Confirm public submission access.

TWO PROJECT WALKTHROUGHS

Supplier decision and a team launch brief

3:30 extended film with slower playback and timed voiceover. Demo Link: Link

LIVE PRODUCT

lightgravity-demo-cr3gb6xbjq-uc.a.run.app

Cloud Run workspace. Share credentials privately.

Google Cloud H2S

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Thank you

